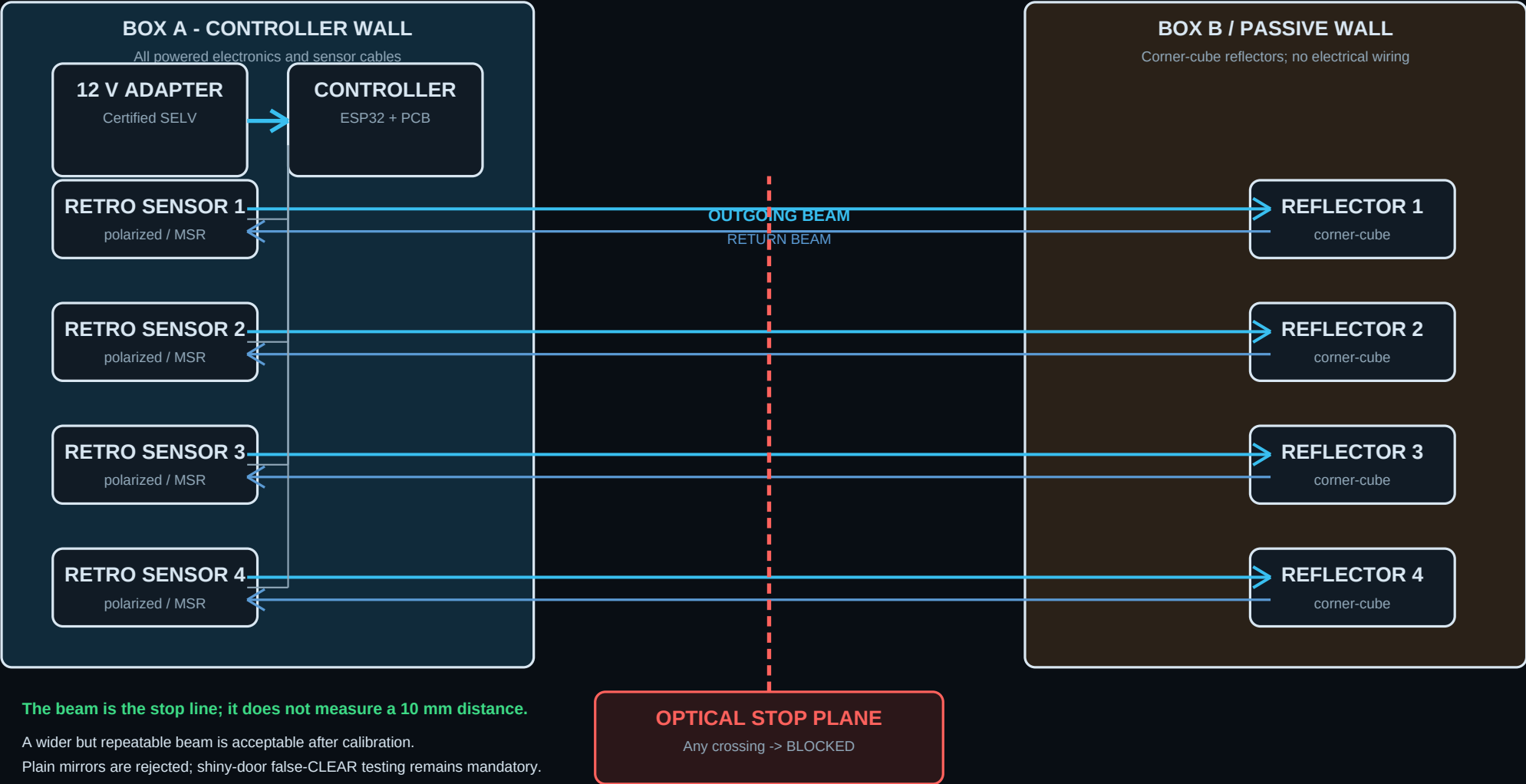
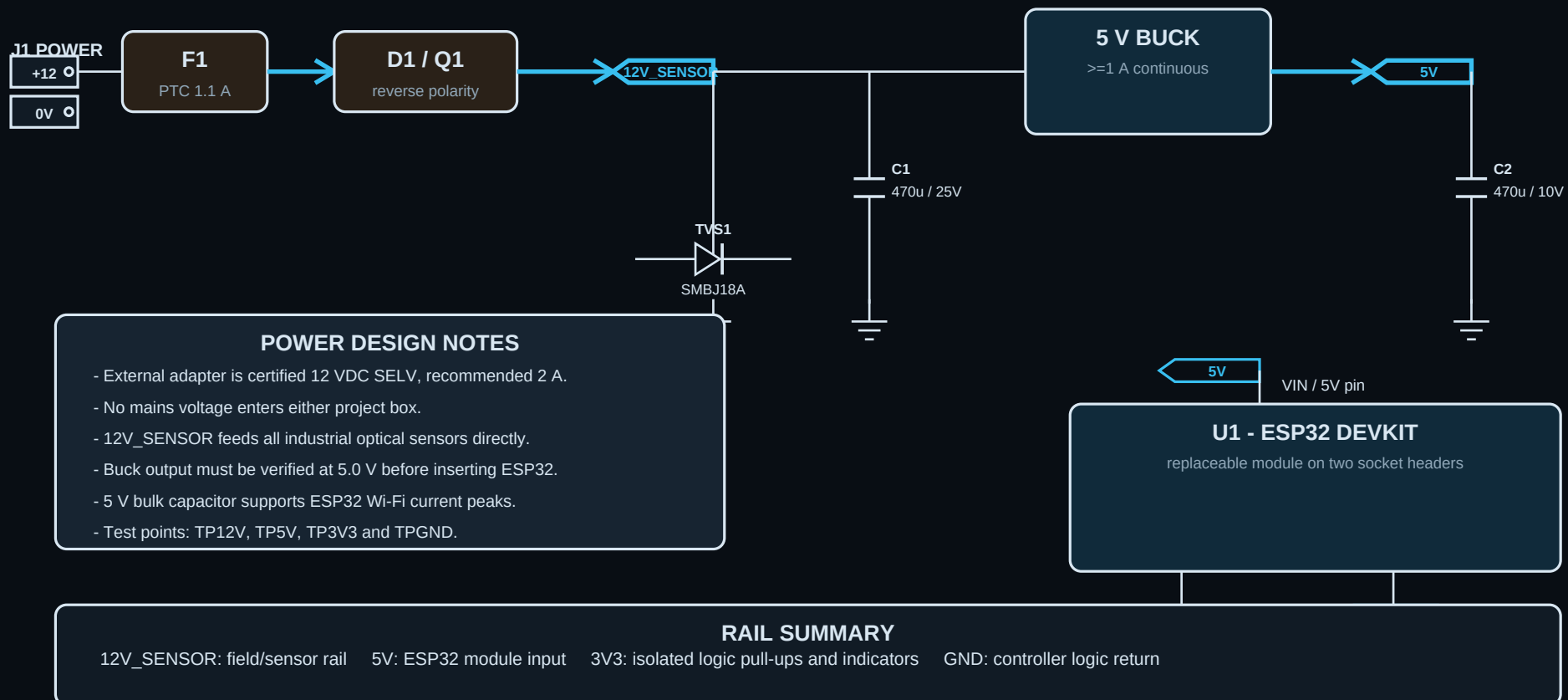


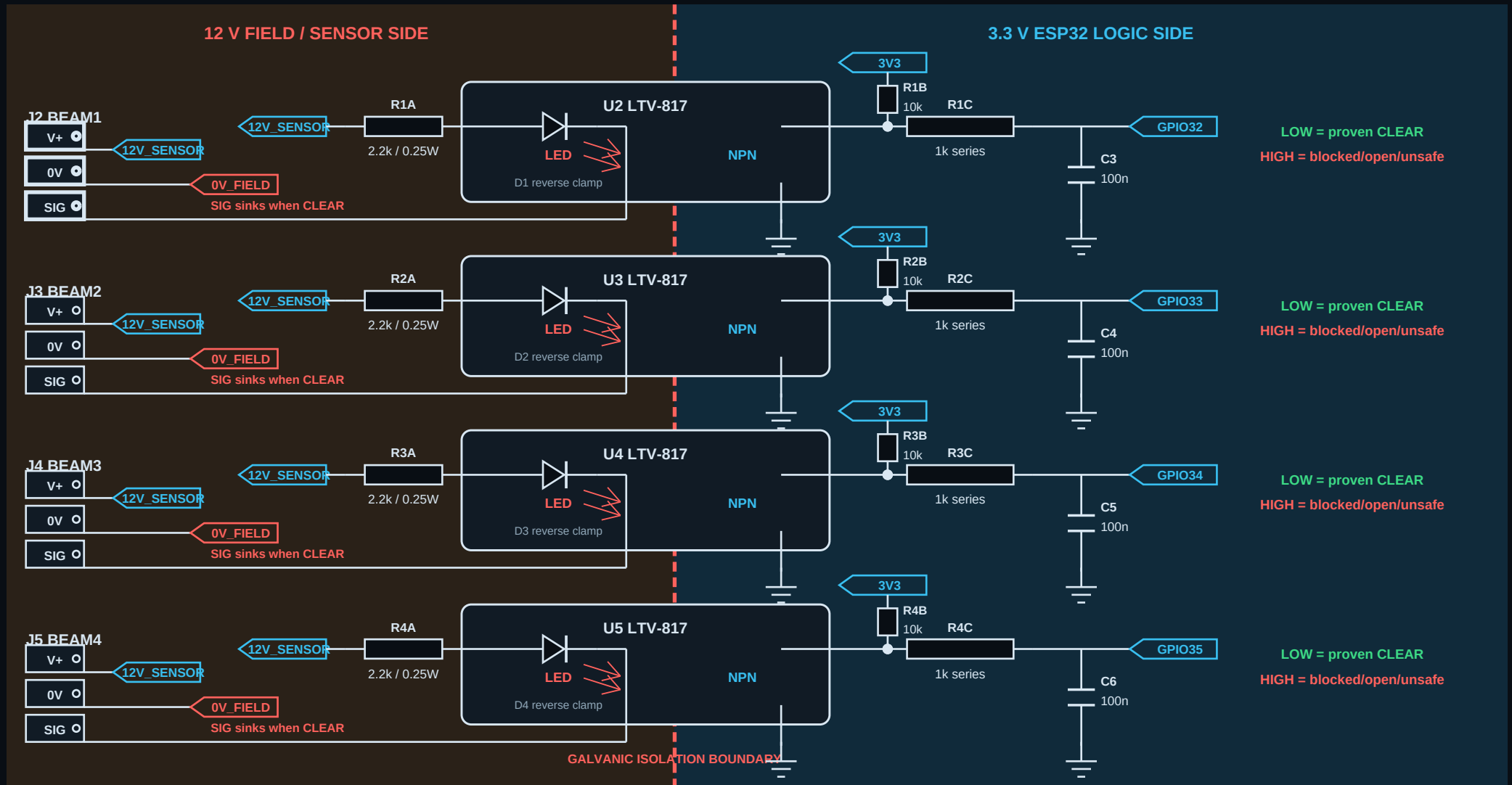
System architecture - fixed polarized retroreflective trip lines



Power input, protected rails and ESP32 module

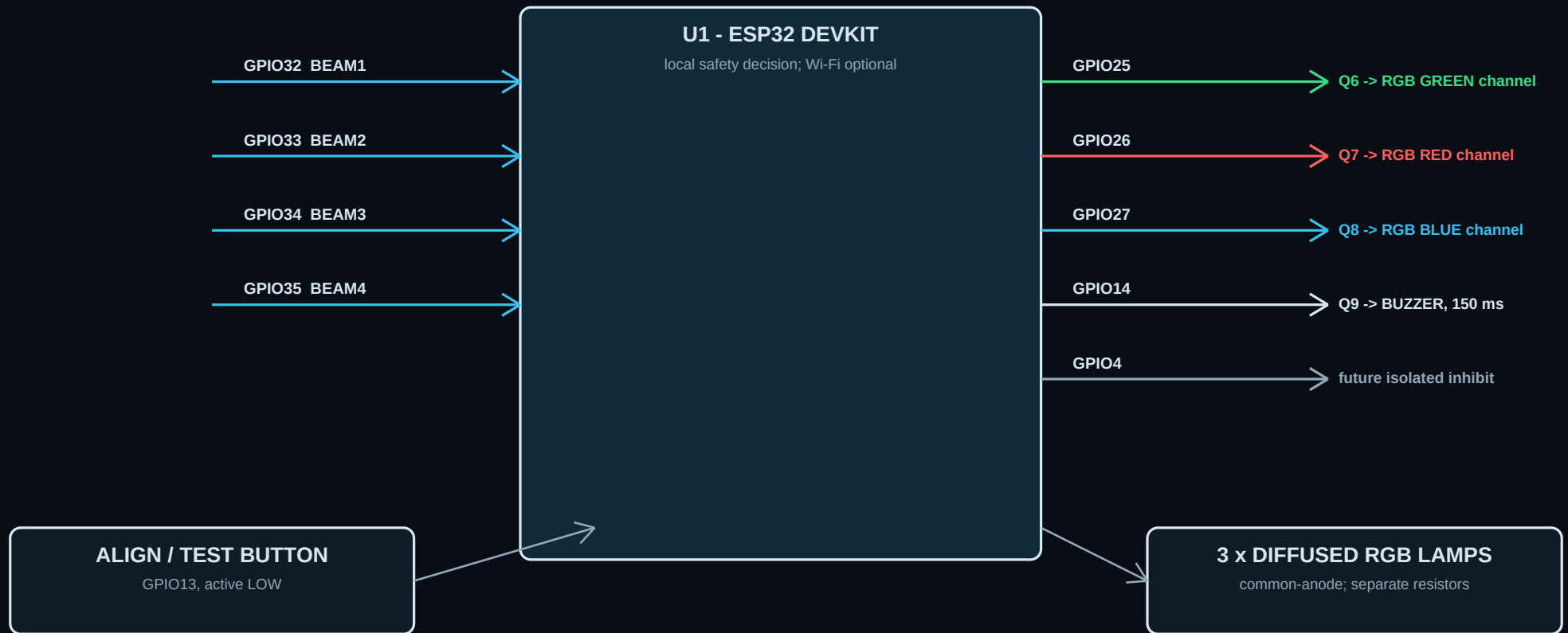


Four isolated fail-safe sensor inputs



D1 behavior: blocked beam, open cable, dead transmitter and dead/open sensor are all the same RED / NOT CLEAR state.

ESP32 connections, redundant RGB indication and safety behavior



GREEN=clear. RED=blocked/no signal/dead sensor. AMBER=boot/unknown (red+green channels). Dark means no power.

Optional RF, door-motion and manual wake system

